AI 3 Month COURSE

📘 DAY 1 – Introduction to Artificial Intelligence

🔹 Artificial Intelligence (AI)

- AI is a field of computer science that enables machines to perform tasks that normally require human intelligence.
- It allows computers to think, learn, and make decisions.

Examples:

- Route prediction in Google Maps by Google
- Face unlock feature in iPhone by Apple
- Chat system like ChatGPT

🔹 Types of AI

- **Narrow AI:** Designed to perform one specific task.
  - Example: YouTube recommendation system
- **General AI:** Can perform any intellectual task like a human.
  - Still in research stage
- **Super AI:** Future concept where machines are smarter than humans.

🔹 Machine Learning (ML)

- Machine Learning is a subset of AI.
- It allows machines to learn from data without being directly programmed.

Examples:

- Fraud detection in banks
- Language translation using Google Translate
- Email spam detection

🔹 **Deep Learning (DL)**

- Deep Learning is a branch of Machine Learning.
- It uses neural networks inspired by the human brain.
- It works well with large and complex data.

Examples:

- Face recognition
- Speech recognition
- Self-driving technology by Tesla

📗 **DAY 2 – Introduction to Shell Scripting**

🔹 **Shell**

- A shell is a program that allows users to interact with the operating system.
- It executes commands entered by the user.

Example: Bash (Bourne Again Shell)

🔹 **Shell Script**

- A shell script is a file that contains multiple Linux commands.
- Commands are executed one by one automatically.

Example:

```
#!/bin/bash
echo "Hello World"
```

🔹 **Variables in Shell**

- Used to store data.
- No spaces around =

- Case sensitive

Example:

```
name="Ali"
echo $name
```

🔹 User Input

- read command is used to take input from user.

Example:

```
echo "Enter your name:"
read username
echo "Hello $username"
```

🔹 Why Shell Scripting is Important

- Automates repetitive tasks
- Saves time
- Used in DevOps and system administration